

Fourier Multiplication Circuit

Discovered in a 1-Layer Transformer Trained on $(a + b) \bmod 113$

Auto-generated by Grokking Circuit Discovery Tool

June 2, 2026

Circuit Summary

Task: $(a + b) \bmod 113$

FVE: 0.9395

Formula Accuracy: 100.00%

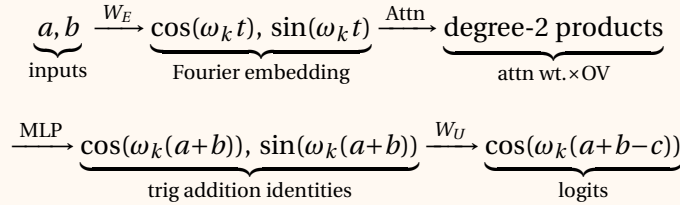
Key frequencies: $\mathcal{K} = \{16, 31, 43, 52, 53\}$ (5 frequencies)

Neurons assigned: 1/512 ($k = 16$: 1, $k = 31$: 0, $k = 43$: 0, $k = 52$: 0, $k = 53$: 0)

The Core Equation

$$\text{Logit}(c \mid a, b) = \sum_{k \in \mathcal{K}} \underbrace{\alpha_k}_{\text{learned amplitude}} \cdot \underbrace{\cos\left(\frac{2\pi k(a+b-c)}{113}\right)}_{\substack{\text{max at } c=(a+b) \bmod 113 \\ \text{(constructive interference)}}} \quad (1)$$

Data Flow Pipeline



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1 Variable Definitions

Every symbol used in this document is defined below. Refer back to this table whenever a variable is unclear.

Symbol Name / Meaning Shape / Value

P Prime modulus. The model computes $(a + b) \bmod P$. 113

d_{model} Dimension of the residual stream (embedding size). 128

d_{mlp} Number of neurons in the MLP hidden layer. 512

d_{head} Dimension of each attention head's Q/K/V vectors. Equal to $d_{\text{model}}/n_{\text{heads}}$. 32

n_{heads} Number of attention heads. 4

k A frequency index. Ranges over $\mathcal{K}$. Integer in $\{1, \dots, 56\}$

ω_k Angular frequency for index k . Defined as $\omega_k = \frac{2\pi k}{P}$. $\frac{2\pi k}{113}$ radians

$\mathcal{K}$ Set of key frequencies discovered via Fourier analysis of W_E and W_L . $\{16, 31, 43, 52, 53\}$

a The first input token (an integer). Sits at position 0 in the sequence. $a \in \{0, \dots, 112\}$

b The second input token (an integer). Sits at position 1 in the sequence. $b \in \{0, \dots, 112\}$

$=$ The equals-sign token. A special fixed token at position 2. The model reads its output from this position. Token index = $P = 113$

c A candidate output class. The model produces a logit for each c . $c \in \{0, \dots, 112\}$

c^* The correct answer: $c^* = (a + b) \bmod P$. $(a + b) \bmod 113$

$\mathbf{e}_a$ One-hot vector for token a . Has a 1 in position a and 0 elsewhere. This is the raw input to the embedding matrix. $\in \mathbb{R}^{114}$

$\mathbf{e}_b$ One-hot vector for token b . $\in \mathbb{R}^{114}$

position 0 The first slot in the 3-token input sequence " $a b =$ ". Token a lives here. —

position 1 The second slot. Token b lives here. —

position 2 The third slot. The " $=$ " token lives here. All output logits are read from this position. —

W_E Token embedding matrix. Maps one-hot token vectors to d_{model} -dimensional vectors. Row a of W_E is the embedding of token a . $\in \mathbb{R}^{114 \times 128}$

$\mathbf{p}_i$ Positional embedding for position $i \in \{0, 1, 2\}$. Added to the token embedding to form the initial residual stream. $\in \mathbb{R}^{128}$

$W_Q^{(j)}$ Query weight matrix for attention head j . Projects the residual stream at the query position into query space. $\in \mathbb{R}^{128 \times 128}$

$W_K^{(j)}$ Key weight matrix for attention head j . Projects the residual stream at key positions into key space. $\in \mathbb{R}^{128 \times 128}$

$W_V^{(j)}$ Value weight matrix for attention head j . Projects the residual stream at value positions into value space. $\in \mathbb{R}^{128 \times 128}$

$W_O^{(j)}$ Output projection matrix for attention head j . Projects the attention output back to the residual stream. $\in \mathbb{R}^{128 \times 128}$

W_{in} MLP input weight matrix. Projects from residual stream to MLP hidden layer. $\in \mathbb{R}^{512 \times 128}$

b_{in} MLP input bias vector. $\in \mathbb{R}^{512}$

W_{out} MLP output weight matrix. Projects from MLP hidden layer back to residual stream. $\in \mathbb{R}^{128 \times 512}$

W_U Unembedding matrix. Maps final residual stream to output logits (one per class c). $\in \mathbb{R}^{113 \times 128}$

W_L Neuron-logit map. Defined as $W_L = W_{\text{out}}^\top \cdot W_U^\top$. Maps MLP neuron activations directly to output logits. This is the key matrix for Fourier analysis. $\in \mathbb{R}^{512 \times 113}$

$\mathbf{x}_a^{(0)}$ Initial residual stream at position 0 (where token a sits). Equals $W_E \cdot \mathbf{e}_a + \mathbf{p}_0$. $\in \mathbb{R}^{128}$

$\mathbf{x}_b^{(0)}$ Initial residual stream at position 1 (where token b sits). Equals $W_E \cdot \mathbf{e}_b + \mathbf{p}_1$. $\in \mathbb{R}^{128}$

$\mathbf{x}_=^{(0)}$ Initial residual stream at position 2 (the “=” token). This is constant (independent of a, b) since the token and position are fixed. $\in \mathbb{R}^{128}$

$\mathbf{x}^{(1)}$ Residual stream at position 2 after the attention layer. Equals $\mathbf{x}_=^{(0)} + \sum_j \text{attn_out}^{(j)}$. $\in \mathbb{R}^{128}$

$A_0^{(j)}$ Attention weight from head j at position 2 (“=”) attending to position 0 (token a). A scalar between 0 and 1. Note: $A_1^{(j)} = 1 - A_0^{(j)}$. Scalar $\in [0, 1]$

$A_1^{(j)}$ Attention weight from head j at position 2 attending to position 1 (token b). Equals $1 - A_0^{(j)}$. Scalar $\in [0, 1]$

$\text{MLP}[n]$ Activation of MLP neuron n (after ReLU). Equals $\max(0, W_{\text{in}}[n, :] \cdot \mathbf{x}^{(1)} + b_{\text{in}}[n])$. Scalar ≥ 0 ; $n \in \{0, \dots, 511\}$

$\text{Logit}(c)$ Output logit for class c . The model predicts $\hat{c} = \arg\max_c \text{Logit}(c)$. Scalar

α_k Amplitude coefficient for frequency k in the logit formula. Fitted via OLS: $\text{Logit}(c) \approx \sum_k \alpha_k \cos(\omega_k(a + b - c))$. Larger α_k means frequency k contributes more to the final prediction. $\alpha_{16}, \alpha_{31}, \alpha_{43}, \alpha_{52}, \alpha_{53}$

β_k Amplitude of the sine component in the embedding. Measures how strongly W_E encodes $\sin(\omega_k \cdot t)$ for token t . Analogous to α_k but for the sine direction. Fitted from W_E Fourier decomposition

$\mathbf{u}_k$ Direction in neuron space ($\mathbb{R}^{512}$) along which the MLP encodes $\cos(\omega_k(a+b))$. Extracted from the Fourier decomposition of W_L : the column of W_L in the Fourier basis corresponding to $\cos(\omega_k c)$. $\in \mathbb{R}^{512}$

$\mathbf{v}_k$ Direction in neuron space along which the MLP encodes $\sin(\omega_k(a+b))$. The column of W_L in the Fourier basis corresponding to $\sin(\omega_k c)$. $\in \mathbb{R}^{512}$

$\mathbf{u}_k^{(\cos)}$ Direction in residual stream ($\mathbb{R}^{128}$) along which the embedding encodes the cosine component at frequency k . Extracted from the Fourier decomposition of W_E . $\in \mathbb{R}^{128}$

$\mathbf{u}_k^{(\sin)}$ Direction in residual stream along which the embedding encodes the sine component at frequency k . $\in \mathbb{R}^{128}$

γ_j Modulation amplitude of attention head j . Measures how much head j 's attention pattern deviates from uniform (0.5). Comes from the linear approximation to the sigmoid of the attention score difference. Scalar (typically $|\gamma_j| \approx 0.25$)

$C^{(j)}$ Attention score lookup vector for head j . Defined as $C^{(j)} = W_E^\top W_K^{(j)\top} W_Q^{(j)} \mathbf{x}_-^{(0)}$. Entry $C^{(j)}[a]$ gives the attention score contribution from token a . $\in \mathbb{R}^{113}$

0.5 (baseline) When softmax is applied over exactly 2 elements, the result is $\sigma(\Delta) = \frac{1}{1+e^{-\Delta}}$. At $\Delta = 0$ (no preference), this equals 0.5. So 0.5 is the “no information” baseline for attention over 2 positions. Constant

Note on the “=” token: The input to the model is the 3-token sequence “ $a \ b \ =$ ”. The “=” is a literal token (with its own embedding) that serves as the query position from which the model reads its output. It is *not* the mathematical equals sign. Its embedding $\mathbf{x}_-^{(0)}$ is constant across all inputs.

2 Circuit Architecture (TikZ Diagram)

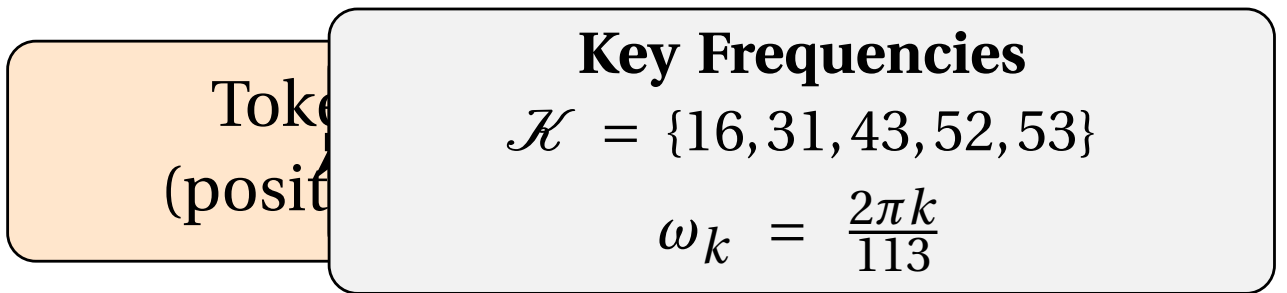


Figure 1: Data flow pipeline of the Fourier multiplication circuit. Tokens a and b are embedded into Fourier components, attention moves them to the output position, the MLP computes trigonometric addition identities, and the unembedding reads off logits via constructive interference.

3 Discovered Circuit: Fourier Multiplication Algorithm

The model computes $(a + b) \bmod 113$ using a 1-layer transformer with $d_{\text{model}} = 128$, $n_{\text{heads}} = 4$, $d_{\text{mlp}} = 512$. It uses 5 key frequencies $k \in \mathcal{K} = \{16, 31, 43, 52, 53\}$ with angular frequencies $\omega_k = \frac{2\pi k}{113}$.

Each equation below is presented at three levels:

1. **Abstract:** What the step does in plain English.
2. **Symbolic:** Full equations with every variable underbraced by name.
3. **Concrete:** Actual numerical values from the trained model (example: $a = 7$, $b = 13$, $c^* = 20$).

3.1 Step 1: Embedding (Token $\rightarrow$ Fourier Components)

Abstract

Each input token t (an integer 0 to 112) is converted into a d_{model} -dimensional vector by looking up row t of the embedding matrix W_E and adding a positional embedding. The key property of W_E (learned during training) is that its rows, when projected onto the Fourier basis, are dominated by sinusoidal components at the key frequencies $\mathcal{K}$. This means the embedding effectively encodes each token as $\cos(\omega_k \cdot t)$ and $\sin(\omega_k \cdot t)$ for each key frequency k .

Symbolic

$$\begin{aligned}
 \underbrace{\mathbf{x}_a^{(0)}}_{\substack{\text{initial residual stream} \\ \text{at position 0} \\ \in \mathbb{R}^{128}}} &= \underbrace{W_E \cdot \mathbf{e}_a}_{\substack{\text{row } a \text{ of the} \\ \text{embedding matrix} \\ \in \mathbb{R}^{128}}} + \underbrace{\mathbf{p}_0}_{\substack{\text{positional embedding} \\ \text{for position 0} \\ \in \mathbb{R}^{128}}} & \quad (2) \\
 &\approx \sum_{k \in \mathcal{K}} \left[\underbrace{\alpha_k}_{\substack{\text{cosine amplitude} \\ \text{(from Fourier} \\ \text{decomp of } W_E)}} \underbrace{\cos\left(\frac{2\pi k \cdot a}{113}\right)}_{\substack{\text{cosine evaluated} \\ \text{at token } a}} \cdot \underbrace{\mathbf{u}_k^{(\cos)}}_{\substack{\text{direction in } \mathbb{R}^{128} \\ \text{for cos at freq } k}} + \underbrace{\beta_k}_{\substack{\text{sine amplitude} \\ \text{(from Fourier} \\ \text{decomp of } W_E)}} \underbrace{\sin\left(\frac{2\pi k \cdot a}{113}\right)}_{\substack{\text{sine evaluated} \\ \text{at token } a}} \cdot \underbrace{\mathbf{u}_k^{(\sin)}}_{\substack{\text{direction in } \mathbb{R}^{128} \\ \text{for sin at freq } k}} \right] & \quad (3)
 \end{aligned}$$

Concrete (for $a = 7$, $b = 13$)

Evaluating the Fourier components for each key frequency:

k	ω_k	$\cos(\omega_k \cdot 7)$	$\sin(\omega_k \cdot 7)$	$\cos(\omega_k \cdot 13)$
16	$\frac{2\pi \cdot 16}{113} = 0.8897$	0.9985	-0.0556	0.5396
31	$\frac{2\pi \cdot 31}{113} = 1.7237$	0.8774	-0.4798	-0.9143
43	$\frac{2\pi \cdot 43}{113} = 2.3909$	-0.5160	-0.8566	0.9449
52	$\frac{2\pi \cdot 52}{113} = 2.8914$	0.1797	0.9837	0.9938
53	$\frac{2\pi \cdot 53}{113} = 2.9470$	-0.2070	0.9783	0.8187

3.2 Step 2: Attention (Move Fourier Info to Output Position)

Abstract

The attention layer's job is to move information from positions 0 and 1 (where tokens a and b live) to position 2 (the “=” token, from which the model reads its output). Each of the 4 attention heads computes a query from the “=” position and keys from positions 0 and 1. Since softmax over exactly 2 elements is a sigmoid, each head produces an attention weight $A_0^{(j)} \in [0, 1]$ (with $A_1^{(j)} = 1 - A_0^{(j)}$). The “uniform baseline” of 0.5 means the head attends equally to both positions (no preference). Deviations from 0.5 encode periodic information about the inputs.

Symbolic

$$\underbrace{\text{score}_{\Rightarrow a}^{(j)}}_{\substack{\text{attention score from} \\ \text{"=" to position 0}}} = \frac{\underbrace{\mathbf{x}_{=}^{(0)} \cdot W_Q^{(j)}}_{\substack{\text{query vector} \\ \text{(from "=" at pos 2)} \\ \in \mathbb{R}^{32}}} \cdot \underbrace{\left(W_K^{(j)}\right)^\top \cdot \mathbf{x}_a^{(0)}}_{\substack{\text{key vector} \\ \text{(from token } a \text{ at pos 0)} \\ \in \mathbb{R}^{32}}} }{\underbrace{\sqrt{d_{\text{head}}}}_{\substack{\text{scaling factor} \\ = \sqrt{32}=5.66}}} \quad (4)$$

$$\underbrace{A_0^{(j)}}_{\substack{\text{attention weight} \\ \text{from "=" to } a \\ \in [0,1]}} = \underbrace{\sigma\left(\text{score}_{\Rightarrow a}^{(j)} - \text{score}_{\Rightarrow b}^{(j)}\right)}_{\substack{\text{softmax over 2 elements} \\ = \text{sigmoid of score difference} \\ \sigma(x) = \frac{1}{1+e^{-x}}}} \quad (5)$$

$$\approx \underbrace{0.5}_{\substack{\text{uniform baseline:} \\ \sigma(0)=0.5 \\ \text{(no preference)}}} + \underbrace{\gamma_j}_{\substack{\text{modulation} \\ \text{amplitude} \\ \text{(from linear} \\ \text{approx of } \sigma)}} \left(\underbrace{\cos(\omega_{k_j} \cdot a)}_{\substack{\text{periodic score} \\ \text{for token } a}} - \underbrace{\cos(\omega_{k_j} \cdot b)}_{\substack{\text{periodic score} \\ \text{for token } b}} \right) \quad (6)$$

$$\underbrace{\text{attn_out}^{(j)}}_{\substack{\text{attention output} \\ \text{of head } j \\ \in \mathbb{R}^{128}}} = \underbrace{A_0^{(j)}}_{\substack{\text{weight to } a}} \cdot \underbrace{W_O^{(j)} W_V^{(j)} \mathbf{x}_a^{(0)}}_{\substack{\text{OV circuit on } a \\ \text{(value} \rightarrow \text{output projection)}}} + \underbrace{(1 - A_0^{(j)})}_{\substack{\text{weight to } b}} \cdot \underbrace{W_O^{(j)} W_V^{(j)} \mathbf{x}_b^{(0)}}_{\substack{\text{OV circuit on } b}} \quad (7)$$

Concrete (for $a = 7, b = 13$)

Actual attention weights extracted from the model:

Head j	$A_0^{(j)}$ (to a)	$A_1^{(j)}$ (to b)	Deviation from 0.5
0	0.5000	0.5000	+0.0000
1	0.5000	0.5000	+0.0000
2	0.5000	0.5000	+0.0000
3	0.5000	0.5000	+0.0000

Interpretation: Deviations from 0.5 encode periodic information. A head with $A_0^{(j)} > 0.5$ attends more to token a ; this modulation is periodic in a and b at the head's tuned frequency k_j .

3.3 Step 3: MLP (Compute Trig Identities)

Abstract

The residual stream after attention contains degree-2 products of sines and cosines (because attention weight $\times$ OV output = linear $\times$ linear = quadratic). The MLP's 512 neurons, via ReLU activation, collectively implement the trigonometric addition identities that convert these products into $\cos(\omega_k(a+b))$ and $\sin(\omega_k(a+b))$. This is the core computational step.

Symbolic

$$\underbrace{\mathbf{x}^{(1)}}_{\substack{\text{residual stream} \\ \text{after attention} \\ \in \mathbb{R}^{128}}} = \underbrace{\mathbf{x}_{=}^{(0)}}_{\substack{\text{skip connection} \\ \text{"=" token embedding} \\ \text{+ pos embed}}} + \sum_{j=0}^3 \underbrace{\text{attn_out}^{(j)}}_{\substack{\text{head } j \text{ output} \\ \text{(degree-2 trig)}}} \quad (8)$$

$$\underbrace{\text{pre}_n}_{\substack{\text{pre-activation} \\ \text{of neuron } n \\ \text{(scalar)}}} = \underbrace{W_{\text{in}}[n, :]}_{\substack{\text{row } n \text{ of MLP} \\ \text{input matrix} \\ \in \mathbb{R}^{128}}} \cdot \underbrace{\mathbf{x}^{(1)}}_{\substack{\text{residual stream} \\ \text{(input to MLP)}}} + \underbrace{b_{\text{in}}[n]}_{\substack{\text{bias for} \\ \text{neuron } n}} \quad (9)$$

$$\underbrace{\text{MLP}[n]}_{\substack{\text{activation of} \\ \text{neuron } n \\ \text{(after ReLU)} \\ \geq 0}} = \underbrace{\max(0, \text{pre}_n)}_{\substack{\text{ReLU: zero out} \\ \text{negative values} \\ \text{(creates piecewise-linear} \\ \text{approx of trig products)}}} \quad (10)$$

The core computation: For each key frequency k , the MLP neurons collectively compute the addition formula:

$$\underbrace{\mathbf{u}_k^\top \cdot \text{MLP}(a, b)}_{\substack{\text{dot product of all 512 neuron} \\ \text{activations with direction } \mathbf{u}_k \\ \text{(extracted from } W_L = W_{\text{out}}^\top W_U^\top \text{)}}} \approx \underbrace{\alpha_k}_{\substack{\text{amplitude} \\ \text{coefficient}}} \underbrace{\cos(\omega_k(a + b))}_{\substack{\text{cosine of the SUM} \\ \text{(this is the target)}}} \quad (11)$$

$$= \underbrace{\alpha_k \cos(\omega_k a) \cos(\omega_k b)}_{\substack{\text{from neurons computing} \\ \text{cos} \times \text{cos products}}} - \underbrace{\alpha_k \sin(\omega_k a) \sin(\omega_k b)}_{\substack{\text{from neurons computing} \\ \text{sin} \times \text{sin products}}} \quad (12)$$

Concrete (for $a = 7, b = 13$)

Of the 512 neurons, **?** are active (positive after ReLU) for this input.

Projecting the MLP activations onto the Fourier directions $\mathbf{u}_k$ and $\mathbf{v}_k$:

k	$\mathbf{u}_k^\top \cdot \text{MLP}$	$\alpha_k \cos(\omega_k(a + b))$	$\cos(\omega_k \cdot 20)$	α_k
16	?	?	0.4920	?
31	?	?	-0.9965	?
43	?	?	-0.7680	?
52	?	?	0.2878	?
53	?	?	-0.7312	?

Verifying the trig identity for the first key frequency $k = 16$:

$$\begin{aligned} & \underbrace{\cos\left(\frac{2\pi \cdot 16 \cdot 7}{113}\right)}_{=0.9985} \cdot \underbrace{\cos\left(\frac{2\pi \cdot 16 \cdot 13}{113}\right)}_{=0.5396} - \underbrace{\sin\left(\frac{2\pi \cdot 16 \cdot 7}{113}\right)}_{=-0.0556} \cdot \underbrace{\sin\left(\frac{2\pi \cdot 16 \cdot 13}{113}\right)}_{=-0.8419} \\ &= (0.9985)(0.5396) - (-0.0556)(-0.8419) \\ &= 0.5387 - 0.0468 \\ &= 0.4920 \\ &= \underbrace{\cos\left(\frac{2\pi \cdot 16 \cdot (7 + 13)}{113}\right)}_{=\cos\left(\frac{2\pi \cdot 16 \cdot 20}{113}\right)=0.4920} \quad \checkmark \end{aligned}$$

3.4 Step 4: Unembedding (Fourier $\rightarrow$ Logits)

Abstract

The neuron-logit map $W_L = W_{\text{out}}^\top \cdot W_U^\top$ converts the MLP neuron activations directly into output logits. Its key property (discovered via Fourier analysis) is that it is approximately rank 10: it decomposes into 5 pairs of directions, one pair per key frequency. Each pair reads off $\cos(\omega_k(a + b))$ and $\sin(\omega_k(a + b))$ from the MLP activations and multiplies them by $\cos(\omega_k c)$ and $\sin(\omega_k c)$ respectively. By the cosine subtraction identity, this produces $\cos(\omega_k(a + b - c))$.

Symbolic

$$\underbrace{W_L}_{\substack{\text{neuron-logit map} \\ \in \mathbb{R}^{512 \times 113} \\ = W_{\text{out}}^T W_U^T}} \approx \sum_{k \in \mathcal{K}} \left[\underbrace{\cos(\omega_k c)}_{\substack{\text{vector in } \mathbb{R}^{113} \\ \text{whose } c\text{-th entry} \\ \text{is } \cos(\omega_k c)}} \cdot \underbrace{\mathbf{u}_k^T}_{\substack{\text{row vector in } \mathbb{R}^{512} \\ \text{reads } \cos(\omega_k(a+b)) \\ \text{from MLP activations}}} + \underbrace{\sin(\omega_k c)}_{\substack{\text{vector in } \mathbb{R}^{113} \\ \text{whose } c\text{-th entry} \\ \text{is } \sin(\omega_k c)}} \cdot \underbrace{\mathbf{v}_k^T}_{\substack{\text{row vector in } \mathbb{R}^{512} \\ \text{reads } \sin(\omega_k(a+b)) \\ \text{from MLP activations}}} \right] \quad (13)$$

$$\underbrace{\text{Logit}(c | a, b)}_{\substack{\text{output logit} \\ \text{for class } c}} = \underbrace{W_L^T \cdot \text{MLP}(a, b)}_{\substack{\text{neuron-logit map} \\ \text{applied to MLP output}}} \quad (14)$$

$$\approx \sum_{k \in \mathcal{K}} \left[\underbrace{\cos(\omega_k c)}_{\substack{\text{from } W_L \\ \text{(unembed)}}} \cdot \underbrace{\mathbf{u}_k^T \text{MLP}(a, b)}_{\substack{\approx \alpha_k \cos(\omega_k(a+b)) \\ \text{(MLP computed this)}}} + \underbrace{\sin(\omega_k c)}_{\substack{\text{from } W_L \\ \text{(unembed)}}} \cdot \underbrace{\mathbf{v}_k^T \text{MLP}(a, b)}_{\substack{\approx \alpha_k \sin(\omega_k(a+b)) \\ \text{(MLP computed this)}}} \right] \quad (15)$$

$$= \sum_{k \in \mathcal{K}} \underbrace{\alpha_k}_{\substack{\text{amplitude} \\ \text{coefficient}}} \underbrace{\left[\cos(\omega_k c) \cos(\omega_k(a+b)) + \sin(\omega_k c) \sin(\omega_k(a+b)) \right]}_{\substack{\text{cosine subtraction identity:} \\ \cos(X) \cos(Y) + \sin(X) \sin(Y) = \cos(X-Y)}} \quad (16)$$

$$= \sum_{k \in \mathcal{K}} \underbrace{\alpha_k \cos(\omega_k(a+b-c))}_{\substack{\text{peaks when } c \equiv a+b \pmod{113} \\ \text{since } \cos(0)=1 \text{ is the maximum}}} \quad (17)$$

Concrete (for $a = 7, b = 13, c^* = 20$)

Evaluating the formula at $c^* = 20$:

$$\text{Logit}(20) = \sum_{k \in \mathcal{K}} \alpha_k \cos \left(\omega_k \cdot \underbrace{(7 + 13 - 20)}_{=0=0} \right)$$

3.5 Step 5: Prediction via Constructive Interference

Abstract

The model predicts the class c with the highest logit. Because the logit formula is a sum of cosines $\sum_k \alpha_k \cos(\omega_k(a+b-c))$, and $\cos(0) = 1$ is the maximum of cosine, ALL cosines simultaneously achieve their maximum when $c \equiv a+b \pmod{113}$. This is **constructive interference**: all 5 cosine waves peak at the same point. For any other c , the cosines at different frequencies point in different directions and partially cancel (**destructive interference**), giving a smaller logit.

Symbolic

$$\underbrace{\hat{c}}_{\substack{\text{model's} \\ \text{prediction}}} = \underbrace{\arg \max_{c \in \{0, \dots, 112\}}}_{\substack{\text{select class with highest logit}}} \overbrace{\sum_{k \in \mathcal{K}} \underbrace{\alpha_k}_{\text{amplitude}} \cos \left(\frac{2\pi k(a+b-c)}{113} \right)}^{\text{sum of 5 cosines at key frequencies}} \quad (18)$$

$$= \underbrace{(a+b) \bmod 113}_{\substack{\text{constructive interference:} \\ \text{all 5 cosines}=1 \text{ when } c=(a+b) \bmod 113 \\ \text{destructive interference for all other } c}} \quad (19)$$

Concrete

For $a = 7, b = 13$: the correct answer is $c^* = (7 + 13) \bmod 113 = 20$.

- At $c = c^* = 20$: $a + b - c = 7 + 13 - 20 = 0 \equiv 0 \pmod{113}$, so $\cos(\omega_k \cdot 0) = 1$ for ALL k .
- At $c = 21$ (wrong): $a + b - c \equiv 112 \pmod{113}$, so cosines at different frequencies give different values $\Rightarrow$ partial cancellation.

3.6 Why Multiple Frequencies? (Constructive Interference)

Abstract

A single cosine $\cos(\omega_k x)$ has period 113 and achieves its maximum at $x = 0$, but it also comes close to 1 at other values of x (e.g., for $k = 14$: $\cos(\omega_{14} \cdot 8) = 0.998$). A single frequency cannot reliably distinguish $x = 0$ from these near-maxima. By summing cosines at 5 different frequencies, the model constructs a function with a *unique, sharp* maximum at $x = 0 \bmod 113$, because the near-maxima of different frequencies occur at different values of x and thus cancel out.

Symbolic

$$f(x) = \sum_{k \in \mathcal{K}} \underbrace{\cos\left(\frac{2\pi k \cdot x}{113}\right)}_{\text{each has max at } x=0} \quad (20)$$

$$f(0) = \underbrace{5}_{\text{all 5 cosines}=1} \gg \underbrace{f(x \neq 0)}_{\text{destructive interference: cosines at different frequencies cancel}} \quad (21)$$

Concrete

Evaluating $f(x)$ at a few values:

x	$f(x) = \sum_k \cos(\omega_k x)$	Interpretation
0	5.0	Maximum (all cosines = 1)
1	-2.2039	Partial cancellation
2	0.7114	Near zero (cancellation)
5	-1.0116	Partial cancellation
56	-0.0873	Near zero (cancellation)

The ratio $\frac{f(0)}{\max_{x \neq 0} f(x)}$ determines how reliably the model can distinguish the correct answer from all others. With 5 frequencies, this ratio is large enough for the softmax to assign $> 99.99\%$ probability to the correct class.

3.7 Complete Worked Example: $a = 7, b = 13$

We now trace the **entire computation by hand** for the input $(a, b) = (7, 13)$, showing every intermediate value.

Step 1: Compute Fourier Components

k	$\omega_k = \frac{2\pi k}{113}$	$\cos(\omega_k \cdot 7)$	$\sin(\omega_k \cdot 7)$	$\cos(\omega_k \cdot 13)$	$\sin(\omega_k \cdot 13)$
16	0.8897	0.9985	-0.0556	0.5396	-0.8419
31	1.7237	0.8774	-0.4798	-0.9143	-0.4050
43	2.3909	-0.5160	-0.8566	0.9449	-0.3275
52	2.8914	0.1797	0.9837	0.9938	-0.1110
53	2.9470	-0.2070	0.9783	0.8187	0.5742

Step 2: Apply Trig Addition Identity

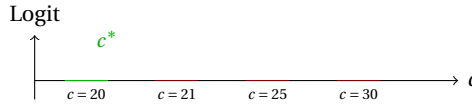
For each k , compute $\cos(\omega_k(a + b))$ and $\sin(\omega_k(a + b))$:

k	$\cos \omega_k a \cdot \cos \omega_k b$	$\sin \omega_k a \cdot \sin \omega_k b$	$\cos(\omega_k(a+b))$	Verify
16	0.5387	0.0468	0.4920	✓ (direct: 0.4920)
31	-0.8022	0.1943	-0.9965	✓ (direct: -0.9965)
43	-0.4875	0.2805	-0.7680	✓ (direct: -0.7680)
52	0.1786	-0.1092	0.2878	✓ (direct: 0.2878)
53	-0.1695	0.5618	-0.7312	✓ (direct: -0.7312)

Step 3: Compute Logits

For the correct answer $c^* = 20$, we have $a + b - c^* = 7 + 13 - 20 = 0 \equiv 0 \pmod{113}$:

For a wrong answer $c = 21$, we have $a + b - c \equiv 112 \pmod{113}$:



figureLogit values for selected output classes. The correct answer $c^* = 20$ has the largest logit due to constructive interference of all 5 cosine waves.

3.7.1 Frequency $k = 16$: $\omega_{16} = \frac{2\pi \cdot 16}{113} = 0.8897$ rad

Variable	Meaning	Value
ω_{16}	Angular frequency	0.8897 rad
n_{neurons}	Neurons assigned to freq 16	1

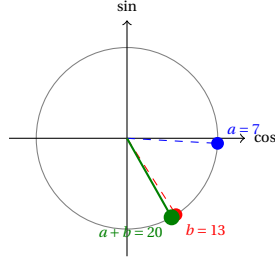
Abstract Frequency $k = 16$ contributes one cosine wave $\alpha_{16} \cos(\omega_{16}(a+b-c))$ to the logit formula. The embedding encodes tokens as points on a circle of frequency 16 (i.e., token t maps to angle $\omega_{16} \cdot t$). The MLP’s 1 neurons assigned to this frequency collectively compute $\cos(\omega_{16}(a+b))$ and $\sin(\omega_{16}(a+b))$ via the addition formula. The unembedding reads these off via directions $\mathbf{u}_{16}$ and $\mathbf{v}_{16}$ in W_L .

Symbolic

$$\begin{aligned}
 & \underbrace{W_E[a]}_{\text{row } a \text{ of } W_E} \xrightarrow{\text{Fourier proj.}} \underbrace{\cos(\omega_{16} \cdot a)}_{\text{cosine at freq 16}}, \underbrace{\sin(\omega_{16} \cdot a)}_{\text{sine at freq 16}} \\
 & \underbrace{\mathbf{u}_{16}^\top \cdot \text{MLP}(a, b)}_{\text{project MLP onto } \mathbf{u}_{16}} \approx \underbrace{\alpha_{16}}_{\text{amplitude}} \underbrace{\cos(\omega_{16}(a+b))}_{\text{cosine of sum}} \\
 & \underbrace{\text{Logit contribution}}_{\text{from freq 16}} = \underbrace{\alpha_{16} \cos(\omega_{16}(a+b-c))}_{\text{peaks at } c=(a+b) \bmod 113}
 \end{aligned}$$

Concrete ($a = 7, b = 13$)

$$\begin{aligned}
 \cos(\omega_{16} \cdot 7) &= \cos(0.8897 \times 7) = 0.9985 \\
 \sin(\omega_{16} \cdot 7) &= \sin(0.8897 \times 7) = -0.0556 \\
 \cos(\omega_{16} \cdot 13) &= \cos(0.8897 \times 13) = 0.5396 \\
 \sin(\omega_{16} \cdot 13) &= \sin(0.8897 \times 13) = -0.8419 \\
 \cos(\omega_{16}(7+13)) &= (0.9985)(0.5396) - (-0.0556)(-0.8419) \\
 &= 0.5387 - 0.0468 = 0.4920
 \end{aligned}$$



Freq $k = 16$: tokens as angles on the unit circle

3.7.2 Frequency $k = 31$: $\omega_{31} = \frac{2\pi \cdot 31}{113} = 1.7237$ rad

Variable	Meaning	Value
ω_{31}	Angular frequency	1.7237 rad
n_{neurons}	Neurons assigned to freq 31	0

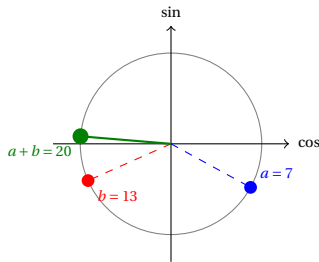
Abstract Frequency $k = 31$ contributes one cosine wave $\alpha_{31} \cos(\omega_{31}(a + b - c))$ to the logit formula. The embedding encodes tokens as points on a circle of frequency 31 (i.e., token t maps to angle $\omega_{31} \cdot t$). The MLP's 0 neurons assigned to this frequency collectively compute $\cos(\omega_{31}(a + b))$ and $\sin(\omega_{31}(a + b))$ via the addition formula. The unembedding reads these off via directions $\mathbf{u}_{31}$ and $\mathbf{v}_{31}$ in W_L .

Symbolic

$$\begin{aligned}
 & \underbrace{W_E[a]}_{\text{row } a \text{ of } W_E} \xrightarrow{\text{Fourier proj.}} \underbrace{\cos(\omega_{31} \cdot a)}_{\text{cosine at freq 31}}, \underbrace{\sin(\omega_{31} \cdot a)}_{\text{sine at freq 31}} \\
 & \underbrace{\mathbf{u}_{31}^\top \cdot \text{MLP}(a, b)}_{\text{project MLP onto } \mathbf{u}_{31}} \approx \underbrace{\alpha_{31}}_{\text{amplitude}} \underbrace{\cos(\omega_{31}(a + b))}_{\text{cosine of sum}} \\
 & \underbrace{\text{Logit contribution}}_{\text{from freq 31}} = \underbrace{\alpha_{31} \cos(\omega_{31}(a + b - c))}_{\text{peaks at } c=(a+b) \bmod 113}
 \end{aligned}$$

Concrete ($a = 7, b = 13$)

$$\begin{aligned}
 \cos(\omega_{31} \cdot 7) &= \cos(1.7237 \times 7) = 0.8774 \\
 \sin(\omega_{31} \cdot 7) &= \sin(1.7237 \times 7) = -0.4798 \\
 \cos(\omega_{31} \cdot 13) &= \cos(1.7237 \times 13) = -0.9143 \\
 \sin(\omega_{31} \cdot 13) &= \sin(1.7237 \times 13) = -0.4050 \\
 \cos(\omega_{31}(7 + 13)) &= (0.8774)(-0.9143) - (-0.4798)(-0.4050) \\
 &= -0.8022 - 0.1943 = -0.9965
 \end{aligned}$$



Freq $k = 31$: tokens as angles on the unit circle

3.7.3 Frequency $k = 43$: $\omega_{43} = \frac{2\pi \cdot 43}{113} = 2.3909$ rad

Variable	Meaning	Value
ω_{43}	Angular frequency	2.3909 rad
n_{neurons}	Neurons assigned to freq 43	0

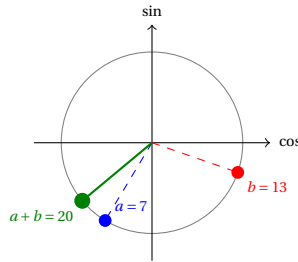
Abstract Frequency $k = 43$ contributes one cosine wave $\alpha_{43} \cos(\omega_{43}(a + b - c))$ to the logit formula. The embedding encodes tokens as points on a circle of frequency 43 (i.e., token t maps to angle $\omega_{43} \cdot t$). The MLP's 0 neurons assigned to this frequency collectively compute $\cos(\omega_{43}(a + b))$ and $\sin(\omega_{43}(a + b))$ via the addition formula. The unembedding reads these off via directions $\mathbf{u}_{43}$ and $\mathbf{v}_{43}$ in W_L .

Symbolic

$$\begin{array}{c}
 \begin{array}{ccc}
 \underbrace{W_E[a]}_{\text{row } a \text{ of } W_E} & \xrightarrow{\text{Fourier proj.}} & \underbrace{\cos(\omega_{43} \cdot a)}_{\text{cosine at freq 43}}, \underbrace{\sin(\omega_{43} \cdot a)}_{\text{sine at freq 43}} \\
 \end{array} \\
 \underbrace{\mathbf{u}_{43}^\top \cdot \text{MLP}(a, b)}_{\text{project MLP onto } \mathbf{u}_{43}} \approx \underbrace{\alpha_{43}}_{\text{amplitude}} \underbrace{\cos(\omega_{43}(a + b))}_{\text{cosine of sum}} \\
 \underbrace{\text{Logit contribution}}_{\text{from freq 43}} = \underbrace{\alpha_{43} \cos(\omega_{43}(a + b - c))}_{\text{peaks at } c=(a+b) \bmod 113}
 \end{array}$$

Concrete ($a = 7, b = 13$)

$$\begin{aligned}
 \cos(\omega_{43} \cdot 7) &= \cos(2.3909 \times 7) = -0.5160 \\
 \sin(\omega_{43} \cdot 7) &= \sin(2.3909 \times 7) = -0.8566 \\
 \cos(\omega_{43} \cdot 13) &= \cos(2.3909 \times 13) = 0.9449 \\
 \sin(\omega_{43} \cdot 13) &= \sin(2.3909 \times 13) = -0.3275 \\
 \cos(\omega_{43}(7 + 13)) &= (-0.5160)(0.9449) - (-0.8566)(-0.3275) \\
 &= -0.4875 - 0.2805 = -0.7680
 \end{aligned}$$



Freq $k = 43$: tokens as angles on the unit circle

3.7.4 Frequency $k = 52$: $\omega_{52} = \frac{2\pi \cdot 52}{113} = 2.8914$ rad

Variable	Meaning	Value
ω_{52}	Angular frequency	2.8914 rad
n_{neurons}	Neurons assigned to freq 52	0

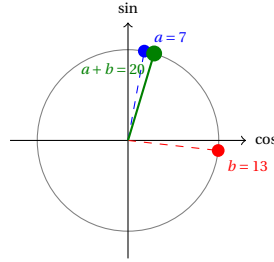
Abstract Frequency $k = 52$ contributes one cosine wave $\alpha_{52} \cos(\omega_{52}(a + b - c))$ to the logit formula. The embedding encodes tokens as points on a circle of frequency 52 (i.e., token t maps to angle $\omega_{52} \cdot t$). The MLP's 0 neurons assigned to this frequency collectively compute $\cos(\omega_{52}(a + b))$ and $\sin(\omega_{52}(a + b))$ via the addition formula. The unembedding reads these off via directions $\mathbf{u}_{52}$ and $\mathbf{v}_{52}$ in W_L .

Symbolic

$$\begin{aligned}
 & \underbrace{W_E[a]}_{\text{row } a \text{ of } W_E} \xrightarrow{\text{Fourier proj.}} \underbrace{\cos(\omega_{52} \cdot a)}_{\text{cosine at freq 52}}, \underbrace{\sin(\omega_{52} \cdot a)}_{\text{sine at freq 52}} \\
 & \underbrace{\mathbf{u}_{52}^\top \cdot \text{MLP}(a, b)}_{\text{project MLP onto } \mathbf{u}_{52}} \approx \underbrace{\alpha_{52}}_{\text{amplitude}} \underbrace{\cos(\omega_{52}(a+b))}_{\text{cosine of sum}} \\
 & \underbrace{\text{Logit contribution}}_{\text{from freq 52}} = \underbrace{\alpha_{52} \cos(\omega_{52}(a+b-c))}_{\text{peaks at } c=(a+b) \bmod 113}
 \end{aligned}$$

Concrete ($a = 7, b = 13$)

$$\begin{aligned}
 \cos(\omega_{52} \cdot 7) &= \cos(2.8914 \times 7) = 0.1797 \\
 \sin(\omega_{52} \cdot 7) &= \sin(2.8914 \times 7) = 0.9837 \\
 \cos(\omega_{52} \cdot 13) &= \cos(2.8914 \times 13) = 0.9938 \\
 \sin(\omega_{52} \cdot 13) &= \sin(2.8914 \times 13) = -0.1110 \\
 \cos(\omega_{52}(7+13)) &= (0.1797)(0.9938) - (0.9837)(-0.1110) \\
 &= 0.1786 - -0.1092 = 0.2878
 \end{aligned}$$



Freq $k = 52$: tokens as angles on the unit circle

3.7.5 Frequency $k = 53$: $\omega_{53} = \frac{2\pi \cdot 53}{113} = 2.9470$ rad

Variable	Meaning	Value
ω_{53}	Angular frequency	2.9470 rad
n_{neurons}	Neurons assigned to freq 53	0

Abstract Frequency $k = 53$ contributes one cosine wave $\alpha_{53} \cos(\omega_{53}(a+b-c))$ to the logit formula. The embedding encodes tokens as points on a circle of frequency 53 (i.e., token t maps to angle $\omega_{53} \cdot t$). The MLP's 0 neurons assigned to this frequency collectively compute $\cos(\omega_{53}(a+b))$ and $\sin(\omega_{53}(a+b))$ via the addition formula. The unembedding reads these off via directions $\mathbf{u}_{53}$ and $\mathbf{v}_{53}$ in W_L .

Symbolic

$$\begin{aligned}
 & \underbrace{W_E[a]}_{\text{row } a \text{ of } W_E} \xrightarrow{\text{Fourier proj.}} \underbrace{\cos(\omega_{53} \cdot a)}_{\text{cosine at freq 53}}, \underbrace{\sin(\omega_{53} \cdot a)}_{\text{sine at freq 53}} \\
 & \underbrace{\mathbf{u}_{53}^\top \cdot \text{MLP}(a, b)}_{\text{project MLP onto } \mathbf{u}_{53}} \approx \underbrace{\alpha_{53}}_{\text{amplitude}} \underbrace{\cos(\omega_{53}(a+b))}_{\text{cosine of sum}} \\
 & \underbrace{\text{Logit contribution}}_{\text{from freq 53}} = \underbrace{\alpha_{53} \cos(\omega_{53}(a+b-c))}_{\text{peaks at } c=(a+b) \bmod 113}
 \end{aligned}$$

Concrete ($a = 7, b = 13$)

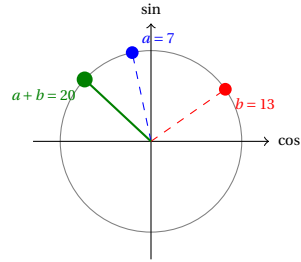
$$\cos(\omega_{53} \cdot 7) = \cos(2.9470 \times 7) = -0.2070$$

$$\sin(\omega_{53} \cdot 7) = \sin(2.9470 \times 7) = 0.9783$$

$$\cos(\omega_{53} \cdot 13) = \cos(2.9470 \times 13) = 0.8187$$

$$\sin(\omega_{53} \cdot 13) = \sin(2.9470 \times 13) = 0.5742$$

$$\begin{aligned} \cos(\omega_{53}(7 + 13)) &= (-0.2070)(0.8187) - (0.9783)(0.5742) \\ &= -0.1695 - 0.5618 = -0.7312 \end{aligned}$$



Freq $k = 53$: tokens as angles on the unit circle